

GoBOLT: Streamlining fleet management and launching a supply-chain SaaS service

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ABOUT CAMIONS

With Google Cloud, Camions Logistics has reduced the cost of fleet management and launched a new cloud-based system that helps customers streamline their supply chains.



About Camions Logistics

Founded in 2015, Camions Logistics aims to lead the Indian logistics sector. Operating as GoBOLT (<https://www.gobolt.in/>), the business provides logistics solutions including express linehaul, planning and order management and warehouse management systems to customers in a wide range of industries. GoBOLT also provides a marketplace platform that gives customers access to more than 5,000 third-party trucks, and the business aims to add logistics-

Google Cloud results

- Reduces the cost of running a business-critical fleet management by 35%
- Helps a client reduce vehicle turnaround times in its yards by 30% with real-time data
- Delivers the capability to expand into international markets such as the United States
- Improves business confidence and capabilities to provide enterprise grade

Reduced
time to
build
application
50%

related services such as brokerage, insurance, resale and enterprise resource planning.

Industries: Transportation & Logistics

Location: India

Google Cloud Deploy.

niveus

About Niveus Solutions

Niveus Solutions is an award-winning Google Cloud partner that drives innovative cloud engineering solutions across India and Singapore. In 2020, it was named Asia-Pacific Breakthrough Google Cloud Partner of the Year.

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Google Cloud (<https://cloud.google.com/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Build (<https://cloud.google.com/build/>)

Google Cloud Deploy
(<https://cloud.google.com/deploy>)

Cloud Functions
(<https://cloud.google.com/functions/>)

Cloud Run (<https://cloud.google.com/run/>)

Cloud Tasks (<https://cloud.google.com/tasks/>)

logistics and
fleet
management
services to
global
brands

Cloud SQL (<https://cloud.google.com/sql/>)

Google Cloud partner that drives

Cloud Firestore

(<https://cloud.google.com/firestore/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

VPC Service Controls

(<https://cloud.google.com/vpc-service-controls>)

Headquartered in New Delhi, India, Camions

Logistics Solutions provides transportation services

under the brand **GoBOLT** (<https://www.gobolt.in/>) to

customers in India and internationally. These

services include full-truckload fulfillment for

multinational and India-based companies, and a

partnership platform that enables smaller

transportation operators to participate in its supply

network.

"Technology and data are critical to us," explains

Aashish Kumar, Chief Technology Officer at GoBOLT.

"We determine where to position our trucks based

on profitability, tracking, and turnaround metrics. We

are building a number of applications to optimize,

scale, and automate our fleet management

operations."

With its truck line haul business mature and well-

established, GoBOLT aims to grow in related areas

including payments, fleet operations, predictive

maintenance and fulfillment optimization.

Finding the right distributed cloud architecture

GoBOLT initially ran its workloads on a traditional cloud service. However, to support expansion into new services and markets such as the United States, the business needed a cloud architecture that was distributed, highly available, fast and scalable.

"We required a cloud-native environment that featured continuous integration and continuous delivery to build and run our applications," explains Kumar. "We also wanted to work with a cloud provider that could support our microservices architecture with a self-healing or automated infrastructure."

Google Cloud (<https://cloud.google.com/>) met these requirements, with Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine>) (GKE) providing a superior managed environment to run containerized applications. With the support of Niveus (<https://niveussolutions.com/>) on day-to-day account management and optimization with Google Cloud, GoBOLT was also able to prioritize the company's needs better for growth.

"Overall, our research determined Google Cloud provided more smart, cloud-native solutions than any other provider, and we decided to move to the service in June 2019," says Kumar.



GoBOLT's supply chain SaaS platform

Optimizing supply chains and fleet management

Shifting to Google Cloud enabled GoBOLT to re-architect the fleet management systems running in its previous cloud environment, and build a supply-chain SaaS platform running in a distributed microservices architecture.

"We envisioned a SaaS platform that would enable us to work with a range of participants, including shippers, brokers and customers, to eliminate their supply-chain bottlenecks," says Kumar.

Meanwhile, the business's preliminary project work found building and running its fleet management systems in Google Cloud could be done at a lower cost than in the previous environment.

After a mandatory lockdown imposed to control the pandemic initially delayed the project, GoBOLT

proceeded with its migration, and by the end of 2020 was running all its fleet management systems in Google Cloud. By running these systems in a containerized environment, the business reduced the management of jobs and services to what Kumar describes as a "trivial" level.

"Using Google Cloud means we can fulfill our mission of launching services quickly, and provide a superior customer experience while decreasing lead times, without investing too much on internal infrastructure firepower," says Kumar.

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Using data to deliver meaningful customer services

GoBOLT runs databases that manage business-critical transaction data, such as shipment orders, in Cloud SQL (<https://cloud.google.com/sql>) to achieve centralization, consistency and high availability. "We designed databases that were consistent and able to handle heavy transaction loads," says Kumar. The business also moved the MongoDB Atlas database used to manage data that powers key applications into Google Cloud.

A BigQuery (<https://cloud.google.com/bigquery>) data warehouse supports GoBOLT's supply chain SaaS platform by powering analytics that customers can use to manage the performance of delivery partners. The service enables customers to answer questions about tracking, freight spend, or even promised versus actual delivery times. They can also view dashboards to determine overall performance.

"We also use BigQuery to underpin a service called 'transporter scorecard' that rates transporters' performance based on profit margin, delivery times, and number of orders fulfilled," says Kumar. "This

enables customers to identify good transport operators in individual regions." He goes on to add that customers can also improve efficiency and performance by reviewing metrics such as loads scheduled for a given week, versus the previous week, to measure throughput.

Delivering the supply chain SaaS platform also required GoBOLT to establish service level objectives and agreements with customers, and Google Cloud provided the tools to help. Ultimately, the business plans to adopt a service mesh to streamline the process of creating applications comprising several microservices, with the native incorporation of service level objectives and agreements through Google Cloud.

"This will help us provide better service and gain greater control over the availability and performance we promise to our customers," says Kumar.

Delivering an enterprise-wide view of data and reducing turnaround times

GoBOLT is building on early market traction achieved by its supply chain SaaS platform by building cloud-native applications in Google Cloud to add new features and services spanning dispatch, procurement, transit and efficiencies in customer service.

For example, Deepak Fertilisers and Petrochemicals Corp. (DFPCL) uses GoBOLT systems to orchestrate its supply chain, spanning inbound and outbound

haulage, as well as primary and secondary activities, which can span from goods and services creation, right up to procurement.

DFPCL also uses GoBOLT's platform to manage the movement of trucks and trailers in yards at a number of plants around India. The real-time, enterprise-wide data delivered by the platform has enabled DFPCL to reduce turnaround times for vehicles in its yards by 30%.

"The data flow is now very collaborative because our platform removed the data silos at DFPCL that had limited the depth and timeliness of the insights available to its managers," says Kumar. "The client started to identify inefficiencies in how it served customers, how it planned transport utilization, its delivery processes and across other scenarios."

DFPCL used a range of matrices to identify and implement efficiencies, which led to the remaining inefficiencies becoming more and more trivial, thanks to GoBOLT's platform. "Because our applications are available all the time, and deliver data and insights in real time, we can provide a supply chain orchestrator workflow that enables customers to model their business edge cases or workflows very easily onto our systems," adds Kumar.



GoBOLT team

Reducing time to build applications by 50%

Implementing continuous integration and continuous deployment through Cloud Build (<https://cloud.google.com/build>) and Cloud Deploy (<https://cloud.google.com/deploy>) has enabled GoBOLT to cut the time needed to build applications by 50%, in comparison to its previous cloud service. "With Google Cloud, it became very easy for us to decrease time-to-market and applications to be live and tested, well before we rolled them out," says Kumar.

With Cloud Build, the business can create builds through a series of custom steps without compromising its security posture, while Cloud Deploy enables continuous deployment to GKE.

Furthermore, the lower cost on Google Cloud when running databases, queueing and network egress

and ingress, combined with its application containerization strategy, has lowered the cost of running the business's fleet management systems by up to 35%.

Google Cloud is also helping accelerate innovation at GoBOLT. A [Firestore](#)

(<https://firebase.google.com/docs/firestore>) NoSQL database manages the IoT data from 400 GoBOLT trucks that ping data back from their GPS systems every three minutes, providing the insights needed for near real-time fleet management. That data feeds into BigQuery to enable GoBOLT to optimize performance and create new service offerings such as SIM-based vehicle tracking.

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Minimizing its infrastructure management load is another priority for GoBOLT and moving into a Virtual Private Cloud (<https://cloud.google.com/vpc>) (VPC) is helping the business achieve its objectives. "There was a perceivable difference when we moved from a similar service on our previous provider into Google Cloud VPC. It was easier to manage as it felt more abstracted than the equivalent service on our alternative provider," says Kumar.

GoBOLT also uses Cloud Functions (<https://cloud.google.com/functions>) to run code without being forced to manage infrastructure, while Cloud Tasks (<https://cloud.google.com/tasks>) enables the business to improve application performance by managing large numbers of distributed tasks.

"Cloud Tasks ensures we can run a highly disconnected architecture," says Kumar. "For example our customer, Hamilton Warehouse, runs auctions for transportation lanes, the transport modes needed to haul consignments between origin

and destination, and plans the vehicle capacity needed, whether that is a large, mid-sized, or small container.

"Hamilton Warehouse also defines the required price. These definitions and requirements are completed the night before the job, to be run early in the morning. The options then go live at the prearranged time and transporters begin bidding for the work.

"This process used to be manual and highly time-consuming. Now, we simply drop a message to Cloud Tasks and this service invokes a series of internal services on Google Cloud at the required time."

Expanding into the United States market

GoBOLT is expanding into the United States, and is working with the Google Cloud team to ensure seamless growth. The business plans to take its supply-chain SaaS software global in early 2023, build on its full truckload fulfillment services by implementing further automation and scaling through the cloud, and acquire and build new applications to support a part-truckload haulage business.

"When you want to deliver on a range of promises, you need to work with a provider that will meet your needs in terms of scalability, performance and time to market," concludes Kumar. "Google Cloud has

given us tremendous confidence that we can achieve our goals."

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